6. Random Variable and Mathematical Expectation

Exercise 6.1

1.

$P(X \le k)$	0.3	0.5	0.9	1

2. $$F_X(x)=\begin{cases}0 & for < 3\\ P_X(3)=0.3 & for\ 3\le x<5\\ P_X(3)+P_X(5)=0.5 & for\ 5\le x<8\\ P_X(3)+P_X(5)+P_X(8)=0.8 & for\ 8\le x<10\\ 1 & for\ x\ge 10\end{cases}$$

5.

$X = x_1$	0	1	2
$P(X = x_1)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

6. (i) 1/10, (ii) 81/100, 19/100, 8/10 (iii) 4

7. Hint: $\int_{-3}^{3} f(x)dx=\int_{-3}^{-1} f(x)dx+\int_{-1}^{1} f(x)dx+\int_{1}^{3} f(x)dx$

8. $(i)\,k=\frac{1}{16},(ii)\ f(x)=\frac{1}{4}(x-1)^3, 1<x\le 3$

9. $A=\frac{1}{5},(i)\frac{1}{e^2},(ii)\frac{e-1}{e},(iii)\frac{e-1}{e^2}$

10. $(a)yes, f(x)=\begin{cases}0, for\ x<0\\ \frac{1}{2}, for\ 0\le x<1\\ 0, for\ 1\le x<2\\ \frac{1}{4}, for\ 2\le x<4\\ 0\ , for\ x\ge 4\end{cases}$ $(b)(i)\frac{1}{4},(ii)\frac{3}{4},(iii)\frac{1}{4}$

Exercise 6.2

1. 3.5 2. 1.8 3. 0.78 4. $\frac{2}{3}$ 5. $\frac{3}{2}, \frac{3}{4}$ 6. ₹ 60

12. Expectation: ₹ 200; Variance: ₹ 21, 60,000; Standard Deviation: ₹ 1,469.69

13. 30 (or 30,000 miles) 14. Expectation: 1; Variance: 9 15. 20

Exercise 6.3

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(c)	(d)	(b)	(c)	(b)	(c)	(d)	(d)	(d)	(d)	(d)	(a)	(c)	(c)	(a)
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
(b)	(b)	(a)	(d)	(b)	(b)	(c)	(c)	(a)	(b)	(c)	(b)	(b)	(b)	(b)

Miscellaneous problems

1. (i) ½ (ii) ¼ (iii) ½ (iv) ¾

2. (a) (i) 13/24 (ii) 0 (b) X is NOT discrete since F is not a step function.

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